BOARD ACTIVITY SHEET: July 2024

Science and Technology Part - 1

Time: 2 Hours **Max. Marks: 40**

Note: i. *All* questions are compulsory.
ii. Use of the calculator is not allowed.
iii. The numbers to the right of the questions indicate full marks.
iv. In case of MCQs (Q. No. 1(A)) only the first attempt will be evaluated and will be given credit.
v. For each MCQ the correct alternative (A), (B), (C), (D) with subquestion number is to be written as an answer.
For Eg. (i) (A) (ii) (B) (iii) (C)
vi. Scientifically and technically correct labelled diagrams should be drawn wherever necessary.

Q.1. (A) Choose the correct alternative: **[5]**

i. In modern periodic table the non-metals are found in _____
(A) s-block (B) p-block
(C) d-block (D) f-block

ii. $CuSO_{4(aq)} + Zn_{(s)} \longrightarrow ZnSO_{4(aq)} + Cu_{(s)}$.
This is _____ type of chemical reaction.
(A) displacement (B) double displacement
(C) combination (D) decomposition

iii. The unit of electric power is _____.
(A) Joule (B) Volt (C) Watt (D) Coulomb

iv. We can see the sun even when it is little below the horizon because of_____:
(A) reflection of light (B) refraction of light
(C) dispersion of light (D) absorption of light

v. _____ is one of the combustible component of L.P.G.
(A) Ethane (B) Propane (C) Methane (D) Ethene

(B) Answer the following: **[5]**

i. Write the correlation.
Brass: Copper and Zinc: : Bronze : _____

ii. Write True or False:
Benzene is an aromatic compound.

iii. Find the odd one out.
Myopia, Presbyopia, Hypermetropia, Retina

iv. What is angle of refraction when angle of incidence is 0°?

v. Name two appliances which work on the principle of magnetic effect of electric current.

Q.2. (A) Give scientific reasons (any *two*): **[4]**

i. Sodium is always kept in kerosene.

ii. A simple microscope is used for watch repairs.

iii. It is necessary to manage the space debris.

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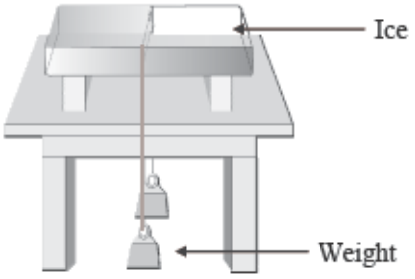
(B) Answer the following questions (any *three*): **[6]**

i. The absolute refractive index of water is 1.33. What is the velocity of light in water? (Velocity of light in vacuum = 3×10^8 m/s.)

ii. Observe the figure and answer the questions given below:



(a) Name the phenomenon shown in the figure.

(b) Define the above phenomenon.

iii.

$$2H_2S + SO_2 \xrightarrow{A} 3S\downarrow + 2H_2O$$

B

(a) Identify the above chemical reaction.

(b) Name the type of reaction in A and B.

iv. Rearrange the column 2 and 3 so as to match with the column 1.

Column 1	Column 2	Column 3
a. Triad	Concentrated mass and +ve charge	Thomson
b. Octave	Average of the atomic masses of first and third element	Newlands
	Properties of the eighth element are similar to first	Dobereiner

v. Define the following terms:

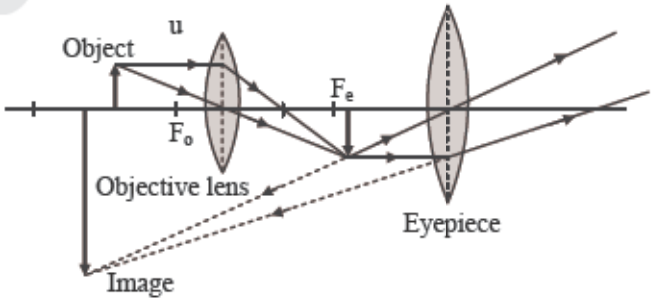
(a) Escape velocity (b) Free fall.

Q.3. Answer the following questions (any *five*): **[15]**

i. What information will you get from the given reaction?

$$CaCO_{3(s)} \xrightarrow{\Delta} CaO_{(s)} + CO_2\uparrow$$

ii. Observe the following ray diagram and answer the questions given below:



(a) Identify the instrument shown in the diagram.

(b) What is the combined effect of two lenses?

(c) Write the use of the above instrument.

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iii. Read the following paragraph and answer the questions given below:

Rainbow is a beautiful natural phenomenon. It is the combined effect of dispersion, refraction and total internal reflection of light. It can be seen mainly after rainfall. Small droplets of water act as small prisms. When light rays from the sun enter these droplets it gets refracted and dispersed. Then there is internal reflection and after that once again the light gets refracted while coming out of the droplet. All these three processes together produce the rainbow.

(a) Which three major phenomena are responsible for rainbow formation?
(b) How does refraction take place?
(c) Draw a diagrammatic representation of rainbow production.

iv. From the elements of second period of the modern periodic table given below, answer the following questions:

3	4	5	6	7	8	9	10
Li	Be	B	C	N	O	F	Ne
Lithium	Beryllium	Boron	Carbon	Nitrogen	Oxygen	Fluorine	Neon
6.941	9.012	10.811	12.011	14.007	15.999	18.998	20.180

(a) Name the elements in which both the shells are completely filled with electrons.
(b) Name the element which has same number of electrons in both the shells.
(c) Which is the most electropositive element in this period?

v. Explain on which factor the value of acceleration due to gravity 'g' depends.

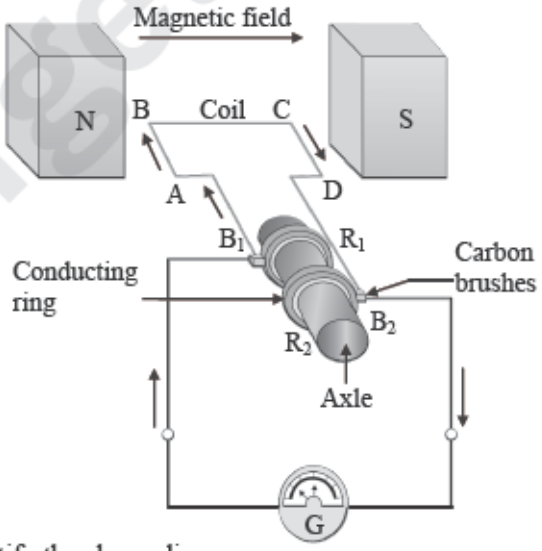
vi. Explain the following terms:
(a) Metallurgy (b) Ores (c) Gangue.

vii. Give the names of the following satellite series developed by India:
(a) Satellite series actively working in the field of telecommunication, television and meteorological services.
(b) Satellite series specially working in the field of education.
(c) Satellite series to exactly locate position of any place on earth's surface.

viii. (a) What is meant by anomalous behaviour of water ?
(b) Explain the role of anomalous behaviour of water in preserving aquatic life in cold climate.

Q.4. Answer the following question (any *one*): **[5]**

i. Observe the following diagram and answer the questions given below :



(a) Identify the above diagram.
(b) Write the principle on which the above appliance works?
(c) Write the working of the above appliance.
(d) Write the use of the above appliance.

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ii. Read the following paragraph and answer the questions given below:

Carbon compounds contain many elements. The element hydrogen is present to a smaller or larger extent in majority of carbon compounds. The compounds which contain carbon and hydrogen as the only two elements are called hydrocarbons. Hydrocarbons are the simplest and the fundamental organic compounds. The smallest hydrocarbon is methane (CH_4), formed by combination of one carbon atom and four hydrogen atoms. We have already seen the structure of methane. Ethane is one more hydrocarbon. Its molecular formula is C_2H_6.

The carbon compounds having a double bond or triple bond between two carbon atoms are called unsaturated compounds. Ethene and ethyne are unsaturated hydrocarbons. The unsaturated hydrocarbons containing a carbon-carbon double bond are called 'Alkenes'. The unsaturated hydrocarbons whose structures contain a carbon-carbon triple bond are called 'Alkynes'. Generally the unsaturated compounds are more reactive than the saturated compounds.

(a) What is meant by hydrocarbon?
(b) Name the smallest hydrocarbon.
(c) Define unsaturated compound.
(d) Differentiate between Alkene and Alkyne.

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